

TIME-DEPENDENT MULTI-MATERIAL FLOW WITH LARGE FLUID DISTORTION

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1. INTRODUCTION

Fluid flow problems in which interfaces between different materials are present are most easily modelled by using a Lagrangian mesh. However, in many 2-D applications distortion of the Lagrangian mesh makes such a method impractical. This paper describes a 2-D, time-dependent, compressible, finite difference Eulerian method which has been developed to simulate such problems.

The change from a Lagrangian to an Eulerian technique introduces two problems (i) tracking interfaces and (ii) advection of fluid properties across cell boundaries. A simple method of representing interfaces is used, in which the portion of each interface in a cell is approximated by a straight line. This technique has been successfully applied to many complex problems in which several interfaces were present. The method is even capable of following thin layers of fluid a fraction of a cell thick. It has been found that the upwind advection method often gives rise to an unacceptable amount of numerical diffusion. The monotonic advection method of Van Leer (1977) has been used to reduce this false diffusion without introducing such defects as spurious oscillations or negative densities.

2. THE LAGRANGIAN AND ADVECTION PHASES

The explicit finite difference (or finite volume) method uses a staggered grid. Pressure (p), density (ρ) and specific internal energy (ϵ) are defined at the centre of each rectangular cell. Components of velocity, u , are defined at cell corners. As in the Los Alamos code FLIC, (Gentry et al, 1966), the Livermore code BBC, (Sutcliffe, 1973), and a number of other codes, the calculation for each time step is divided into two phases.

The first phase may be thought of as a Lagrangian step in which the mesh moves with the fluid. In this phase the changes in velocity and internal energy due to the pressure terms are calculated. The equations integrated are

$$\rho \frac{\partial \underline{u}}{\partial t} = -\nabla p \text{ and } \rho \frac{\partial \epsilon}{\partial t} = -p \operatorname{div} \underline{u}.$$

In the Lagrangian phase both spatial directions X and Y are calculated simultaneously.

In the second advection phase, transport of mass, internal energy and momentum across cell boundaries is computed. This may be thought of as remapping the displaced mesh at the end of phase 1 back to its original position. X and Y advection are calculated in separate steps. The order of calculation is X first, Y second and Y first, X second for alternate time steps. XY splitting greatly simplifies the interface tracking procedure and enables the 1-D advection method of Van Leer (1977) to be used.

For low Mach number flows the Lagrangian phase is divided into several, say L, steps each of which is explicit and second order accurate in time. This significantly improves the efficiency of the method as fewer of the more complex advection steps then need to be performed. The over-all time step, Δt must meet the following restrictions

$$\Delta t < L \frac{\min(\Delta x, \Delta y)}{c}, \frac{\Delta x}{|u|}, \frac{\Delta y}{|v|}$$

where c is the sound speed and u,v are the velocity components.

3. TREATMENT OF MATERIAL INTERFACES

A possible way of tracking interfaces would be to define each interface by a set of Lagrangian marker particles. However, this method becomes logically complicated if interfaces become highly distorted or if the geometry is complex. The author prefers to use a simpler method in which the material layout is described solely by the volume fractions of each fluid in each cell. The first method of this type to be widely used was the Donor-Acceptor Cell technique, variants of which are described by Ramshaw and Trapp (1976) and Hirt and Nichols (1981). Interfaces are not precisely located within each cell, but transport across cell sides is calculated by a method which inhibits different materials diffusing into one another. In a similar method, the SLIC technique of Noh and Woodward (1976), for the

purpose of calculating transport across cell sides, interfaces are drawn parallel to either the X or Y axes.

The method used here represents interfaces more accurately. The portion of an interface in a cell is approximated by a straight line. The method of locating this straight line for the case of a cell containing two materials A and B is shown in Fig. 1(a) (b) (c) and (d). Volume fractions of material A for the cell under consideration and its eight surrounding cells are used to determine the slope of the interface. The position of the interface is then adjusted so that it divides the cell into two areas which match the two volume fractions. During this calculation the cell is treated as a square of size 1 unit $\times$ 1 unit.

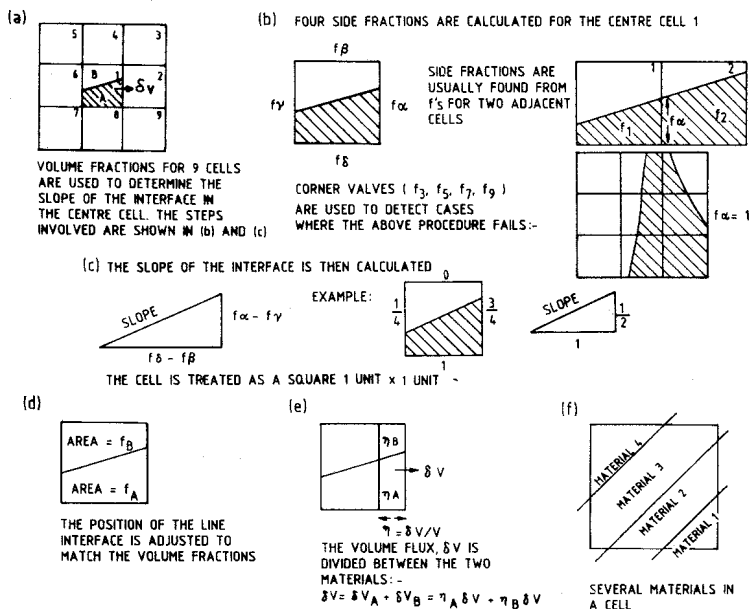


Fig. 1 The interface treatment.

As X-advection and Y-advection are calculated in separate steps it is sufficient to consider the flow across one side only. The interface position is used to divide the volume of fluid, δV , flowing across a cell side, between the two fluids, as indicated in Fig. 1(e),

$$\delta V = \delta V_A + \delta V_B$$

The corresponding mass and internal energy fluxes are then

$$\delta m_A = \rho_A \delta V_A, \quad \delta m_B = \rho_B \delta V_B, \quad \delta E_A = \epsilon_A \delta m_A, \quad \delta E_B = \epsilon_B \delta m_B$$

where ρ_A , ρ_B , ϵ_A and ϵ_B are suitably defined density and internal energy values. During the X and Y advection steps the total mass and total internal energy of each material are conserved.

The portions of an interface in each cell are located independently and are not forced to join up. However, as illustrated by the results in section 5, near continuous interfaces are obtained unless structures of order the mesh size develop. The ability to treat each mixed material cell independently in this manner is one of the reasons for the simplicity and effectiveness of the method.

A cell which contains several interfaces is treated by repeated application of the two-component (A and B) technique. For the example shown in Fig. 1(f) the calculation proceeds as follows:

- (a) A is material 1, B is material 2 + 3 + 4: the interface between materials 1 and 2 is defined.
- (b) A is material 1 + 2, B is material 3 + 4: the interface between materials 2 and 3 is defined.
- (c) A is material 1 + 2 + 3, B is material 4: the interface between materials 3 and 4 is defined.

It is necessary for the user of the program to specify the order in which layers of material are present. Use of this technique enables thin layers of fluid less than a cell thick to be followed accurately.

4. THE ADVECTION METHOD

The application of the method used in the hydrocode to the simple one dimensional problem,

$$\frac{\partial \rho}{\partial t} + u \frac{\partial \rho}{\partial x} = 0 \quad \text{where } u = \text{a positive constant,}$$

is described first. As the advection phase is split into two steps X and Y, the extension to two dimensions is not difficult. The difference approximation used for the above equation has the following conservation form:

$$\frac{\rho_{j+\frac{1}{2}}^{n+1} - \rho_{j+\frac{1}{2}}^n}{\Delta t} + \frac{\bar{\rho}_{j+1} - \bar{\rho}_j}{\Delta x} = 0.$$

The mass flowing across cell boundary j during the time step is $\bar{\rho}_j u \Delta t$, where the density value is given by

$$\bar{\rho}_j = \rho_{j-\frac{1}{2}}^n + \frac{1}{2} (1 - \eta) \Delta x D_j, \text{ with } \eta = \frac{u \Delta t}{\Delta x}.$$

$\rho_{j-\frac{1}{2}}^n$ is the density value in the upwind cell at the beginning of the time step. The term containing $D_j \approx \partial \rho / \partial x$ allows for the density gradient within the upwind cell. Possible choices for D_j are

First order: $D_j = 0$ (the upwind method)

$$\text{Second order: } D_j = \frac{\rho_{j+\frac{1}{2}}^n - \rho_{j-\frac{1}{2}}^n}{\Delta x}$$

$$\text{Third order: } D_j = \frac{2 - \eta}{3} \frac{\rho_{j+\frac{1}{2}}^n - \rho_{j-\frac{1}{2}}^n}{\Delta x} + \frac{1 + \eta}{3} \frac{\rho_{j-\frac{1}{2}}^n - \rho_{j-\frac{3}{2}}^n}{\Delta x}.$$

The numerical diffusion present in the first order method often gives poor results. This diffusion is reduced in the second or third order methods but non-physical oscillations and negative densities may then occur. The third order method, which has less numerical dispersion, gives rather better results than the second order method. The non-physical behaviour is eliminated if the monotonic advection methods of Van Leer (1977) are used. In these methods, non-linear cut-offs applied to D_j ensure that the new density value $\rho_{j+\frac{1}{2}}^{n+1}$ satisfies the following monotonicity conditions:

$$\min(\rho_{j-\frac{1}{2}}^n, \rho_{j+\frac{1}{2}}^n) \leq \rho_{j+\frac{1}{2}}^{n+1} \leq \max(\rho_{j-\frac{1}{2}}^n, \rho_{j+\frac{1}{2}}^n).$$

The formula used here for D_j is

$$D_j \Delta x = S \min\{\bar{D}_j \Delta x, 2|\rho_{j+\frac{1}{2}}^n - \rho_{j-\frac{1}{2}}^n|, 2|\rho_{j-\frac{1}{2}}^n - \rho_{j-\frac{3}{2}}^n|\}$$

$$\text{where } S = \begin{cases} 0 & \text{if } \text{sign}(\rho_{j+\frac{1}{2}}^n - \rho_{j-\frac{1}{2}}^n) \neq \text{sign}(\rho_{j-\frac{1}{2}}^n - \rho_{j-\frac{3}{2}}^n) \\ \text{sign}(\rho_{j+\frac{1}{2}}^n - \rho_{j-\frac{1}{2}}^n) & \text{otherwise.} \end{cases}$$

The standard advection method in the hydrocode uses the above third order approximation for $\bar{D}_j$. Alternative methods for calculating D_j are described in the recent paper (Van Albada et al. 1982).

Results for a test problem, the advection of a square pulse through 300 meshes, are shown in Fig. 2. The calculations used 100 meshes in the x-direction with periodic boundary conditions. η was equal to 0.1.

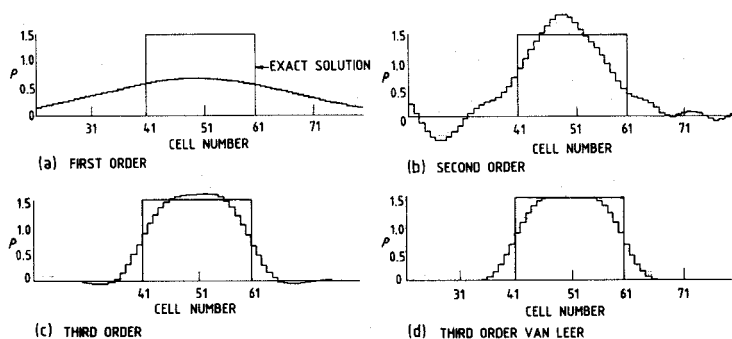


Fig. 2 Advection of a square pulse with uniform velocity.

The Van Leer advection method is used in the hydrocode at the end of the Lagrangian phase. The values of density, internal energy and velocity components at this stage are denoted by

$$\tilde{\rho}_{j+\frac{1}{2}k+\frac{1}{2}}^n = \rho_{j+\frac{1}{2}k+\frac{1}{2}}^n V_{j+\frac{1}{2}k+\frac{1}{2}}^n / \tilde{V}_{j+\frac{1}{2}k+\frac{1}{2}}^n, \quad \tilde{\epsilon}_{j+\frac{1}{2}k+\frac{1}{2}}^n, \quad \tilde{u}_{jk}^n \text{ and } \tilde{v}_{jk}^n.$$

The density value allows for the change in cell volume, $V_{j+\frac{1}{2}k+\frac{1}{2}}^n$ to $\tilde{V}_{j+\frac{1}{2}k+\frac{1}{2}}^n$, during the Lagrange phase. For the case when X-advection is calculated first and Y-advection is calculated second, $\tilde{\rho}_{j+\frac{1}{2}k+\frac{1}{2}}^n$ is updated to $\rho_{j+\frac{1}{2}k+\frac{1}{2}}^*$ during the X-step and $\rho_{j+\frac{1}{2}k+\frac{1}{2}}^*$ is updated to $\rho_{j+\frac{1}{2}k+\frac{1}{2}}^{n+1}$ during the Y-step. The Van Leer method uses three values of $\tilde{\rho}^n$ to calculate the flux across a cell side in the X-step and three values of ρ^* for a Y-step flux. If u and v are positive, the updated densities satisfy the following monotonicity conditions.

For the X-step

$$\min(\tilde{\rho}_{j-\frac{1}{2} k+\frac{1}{2}}^n, \tilde{\rho}_{j+\frac{1}{2} k+\frac{1}{2}}^n) \leq \rho_{j+\frac{1}{2} k+\frac{1}{2}}^* \leq \max(\tilde{\rho}_{j-\frac{1}{2} k+\frac{1}{2}}^n, \tilde{\rho}_{j+\frac{1}{2} k+\frac{1}{2}}^n)$$

and for the Y-step

$$\min(\rho_{j+\frac{1}{2} k-\frac{1}{2}}^*, \rho_{j+\frac{1}{2} k+\frac{1}{2}}^*) \leq \rho_{j+\frac{1}{2} k+\frac{1}{2}}^{n+1} \leq \max(\rho_{j+\frac{1}{2} k-\frac{1}{2}}^*, \rho_{j+\frac{1}{2} k+\frac{1}{2}}^*).$$

Then for the advection phase as a whole $\rho_{j+\frac{1}{2} k+\frac{1}{2}}^{n+1}$ lies between the maximum and minimum of $\tilde{\rho}_{j-\frac{1}{2} k-\frac{1}{2}}^n, \tilde{\rho}_{j-\frac{1}{2} k+\frac{1}{2}}^n, \tilde{\rho}_{j+\frac{1}{2} k-\frac{1}{2}}^n, \tilde{\rho}_{j+\frac{1}{2} k+\frac{1}{2}}^n$. Similar results hold for different flow directions and for the variables $\epsilon_{j+\frac{1}{2} k+\frac{1}{2}}^{n+1}, u_{jk}^{n+1}$ and v_{jk}^{n+1} .

Experience in using the hydrocode for a wide range of applications has led to the following conclusions concerning the choice of advection method.

For advection of mass and internal energy the upwind method is usually adequate if variations of ρ and ϵ within each material are gradual. The interface treatment preserves the discontinuities in ρ and ϵ at material boundaries. If steep gradients of ρ or ϵ are present within a material then the Van Leer method is the best choice.

If the upwind method is used for momentum advection the numerical diffusion introduced dissipates kinetic energy and as a result distortion of material interfaces may be inhibited. An improved momentum advection method has often been found necessary. The Van Leer method greatly reduces the dissipation without producing an unstable solution.

Many of these points are illustrated in the following applications.

5. APPLICATIONS

5.1 Rayleigh-Taylor instability

In the starting configuration for this problem a fluid with $\rho = 2$ rests on top of a fluid with $\rho = 1$. Gravity acts downwards. The interface between the two fluids is initially horizontal but a velocity perturbation is present. As a result a spike of heavy fluid falls and bubbles of light fluid rise. Material properties have been chosen to give a low Mach number

flow. The behaviour is then similar to that calculated for incompressible Rayleigh-Taylor instability, see (Daly, 1967).

Calculations have been performed in planar geometry using 40 zones in the horizontal direction and 100 zones in the vertical direction. For this application benefit is gained by using several, 4 in this case, Lagrange steps per advection step. An over-all time step of $\Delta t = 3\Delta x/c$ was used without loss of stability (shown by the pressure contours in Fig. 3(c)).

Figs. 3(a) and (b) show the effect of changing the momentum advection method. The vortices which are formed as the spike of denser fluid falls are partially damped out and the roll-up of the interface is reduced if the upwind method is used.

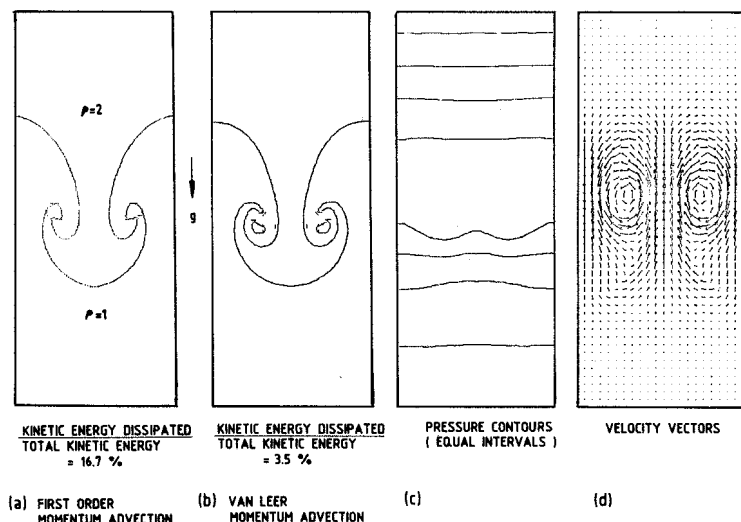


Fig. 3 Rayleigh-Taylor instability.

5.2 A shock tube problem

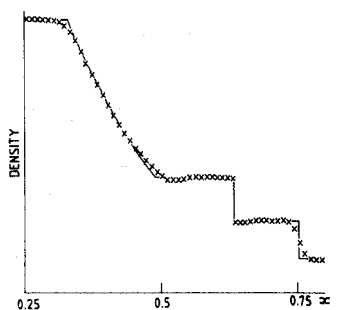
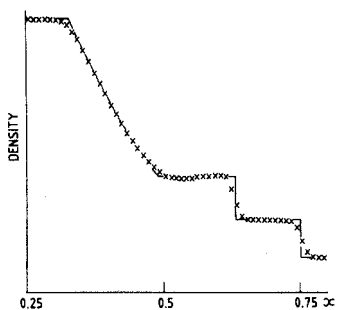
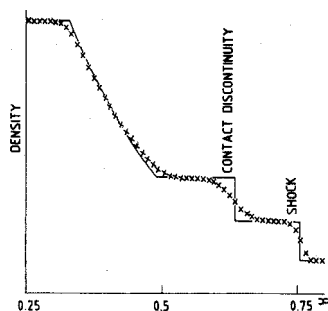
Calculations have been performed on the one-dimensional shock tube problem studied by Sod (1977). In Figs. 4(a) and 4(b) density profiles are shown for single material calculations. There is considerable smearing of the contact discontinuity when the upwind method is used for mass advection. The Van Leer advection method gives a large improvement. Alternatively the contact discontinuity may be treated as a material interface as illustrated in Fig. 4(c). Smearing of the contact discontinuity is then eliminated completely. As the methods of Van Leer (1979)

$p = 1$	$p = 0.1$
$\rho = 1$	$\rho = 0.125$
$u = 0$	$u = 0$
$\epsilon = 2.5$	$\epsilon = 2.0$

PERFECT GAS WITH $\gamma = 1.4$ PERFECT GAS WITH $\gamma = 1.4$

$x = 0.5$

(a) THE INITIAL CONDITIONS



DISTANCE MOVED BY
CONTACT DISCONTINUITY = 13.36 ZONES (EXACT)
= 13.31 ZONES (CALCULATED)

Fig. 4 Results for the shock tube problem of Sod (1978).

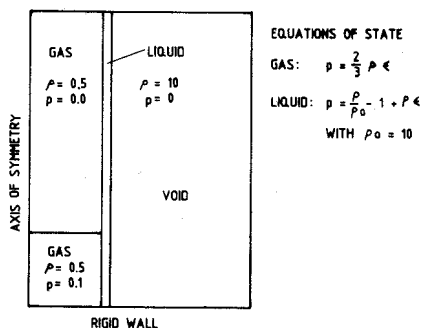
have not been used in the Lagrangian phase smearing of the shock front is still present.

5.3 An expanding pipe

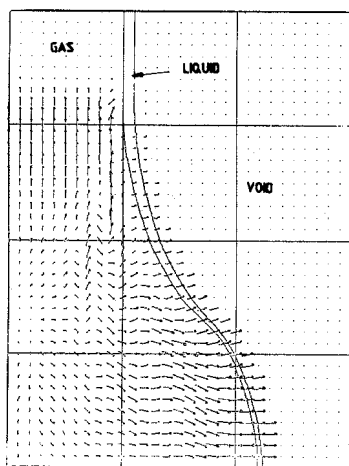
This example demonstrates the ability of the interface tracking method to follow thin layers of fluid. A cylindrical pipe of liquid with $\rho = 10$ is filled with gas at $\rho = \frac{1}{2}$. Initially the gas at one end is pressurised. During the course of the calculation a shock wave travels through the gas and the pipe expands.

Interface and velocity vector plots are shown in Fig. 5. At the time illustrated the expanded end of the pipe is less than half a cell thick. The position of the expanded pipe calculated by the Eulerian code was checked against results from a

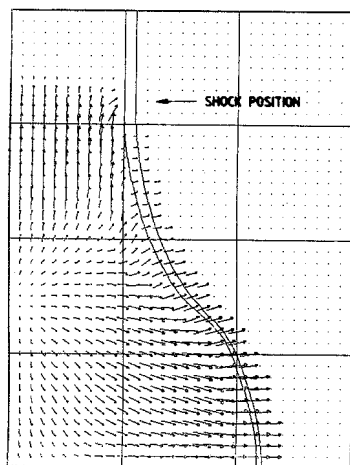
Lagrangian code and found to be in good agreement. If second order momentum advection is used, without the cut-offs on D_j proposed by Van Leer, an oscillatory velocity field is obtained. These supposed spurious effects are eliminated if the Van Leer method is used.



(a) THE INITIAL CONFIGURATION



(b) SECOND ORDER MOMENTUM ADVECTION



(c) VAN LEER MOMENTUM ADVECTION

Fig. 5 Propagation of a shock wave inside an expanding pipe.

5.4 Explosively driven flows

Results are given here for application of the multi-material hydrocode to two experiments which were performed at AWRE.

In the first experiment a small explosive charge was detonated inside a brass cylinder, as illustrated in Fig. 6(a). The expansion of the outer surface of the cylinder in the experiment was obtained by high-speed photography. The calculation used

72 $\times$ 88 meshes to cover one quadrant of the flow. At the earlier time shown in Fig. 6 the observed interface position agrees well with the computer prediction. At the later time, the experiment lags slightly behind the calculation. This may be because the metal cylinder started to fracture by this time.

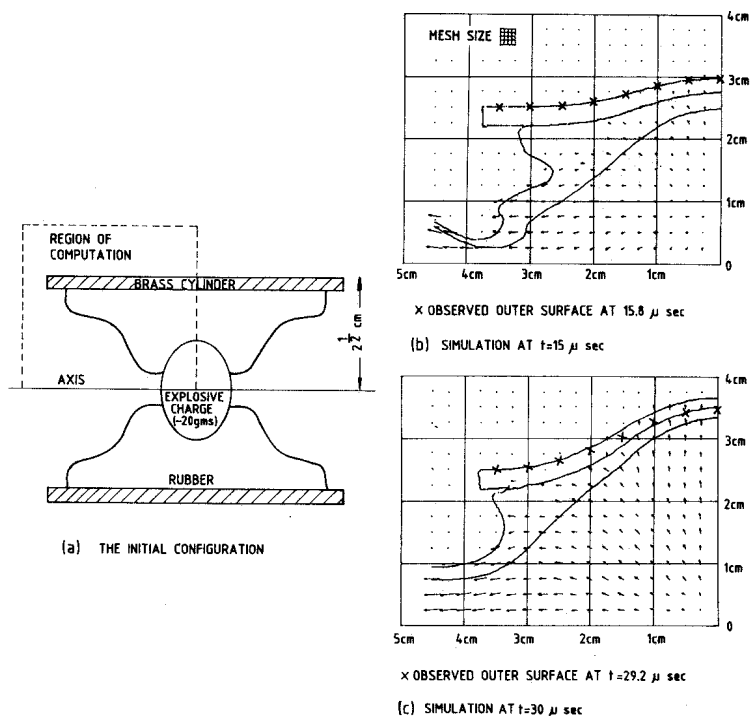


Fig. 6 Brass cylinder experiment.

In the second application the hydrocode is used to calculate a shaped charge experiment. High explosive drive is applied to a conical liner. As the liner collapses a high-velocity jet forms on the axis of symmetry. The computational region used 40 $\times$ 200 meshes as indicated in Fig. 7 and the calculation took approximately 10 minutes on a CRAY-1 computer. The interface plots in Fig. 7 show the formation of the jet and the late-time behaviour of the collapsed liner. Comparison with an X-ray of the experiment shows that the computed and observed liner shapes agree reasonably well at a late time.

These two applications show the advantages of the present Eulerian method over the conventional Lagrangian approach to

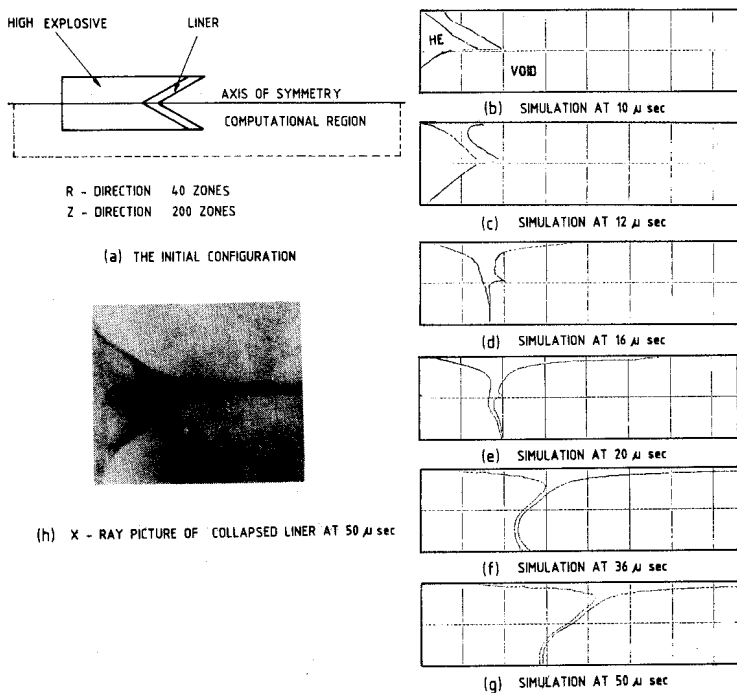


Fig. 7 Shaped charge experiment.

multimaterial flow. If the Eulerian method is used calculations may be continued to a very late stage and still give satisfactory results.

ACKNOWLEDGEMENTS

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